

A PROJECT TITLE

ON

"Multi-Class Disease Diagnosis Using Multinomial Logistic
Regression"

project report submitted in partial fulfilment of the requirements for the Course

**MACHINE LEARNING
BACHELOR OF TECHNOLOGY**

In

A.Y.: 2026-27

by

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Name of Title: "Multi-Class Disease Diagnosis Using Multinomial Logistic Regression"

1 . Abstract:

The increasing demand for accurate and timely disease diagnosis has encouraged the use of Machine Learning techniques in the healthcare sector. This project, "**Multi-Class Disease Diagnosis Using Multinomial Logistic Regression**," aims to develop an intelligent system that predicts multiple diseases based on patient symptoms and clinical information. The proposed model uses the Multinomial Logistic Regression algorithm, which is suitable for multi-class classification problems and can identify one disease from several possible disease categories.

The healthcare dataset is preprocessed by handling missing values, encoding categorical variables, selecting relevant features, and normalizing the data to improve model performance. The processed dataset is then used to train and test the machine learning model. After training, the system predicts the most probable disease when new patient information is provided. The model performance is evaluated using standard metrics such as accuracy, precision, recall, F1-score, and confusion matrix.

The proposed system provides a simple and user-friendly interface that enables quick disease prediction, supporting healthcare professionals in making preliminary diagnostic decisions. Although it is not intended to replace medical experts, it serves as an effective decision-support tool that reduces diagnosis time and improves prediction accuracy. This project demonstrates the practical application of machine learning in healthcare and highlights its potential to enhance clinical decision-making and patient care.



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2.Introduction:

Healthcare is one of the most important sectors where accurate and timely disease diagnosis plays a vital role in improving patient care. Hospitals and healthcare organizations generate large volumes of patient data, including symptoms, medical histories, laboratory reports, and diagnostic records. Analyzing this data manually can be time-consuming and may lead to errors. Machine Learning (ML) techniques provide an effective solution by identifying patterns in healthcare data and assisting medical professionals in making faster and more accurate diagnostic decisions.

This project focuses on developing a Multi-Class Disease Diagnosis System using the Multinomial Logistic Regression algorithm. The objective is to predict one disease from multiple possible disease categories based on patient symptoms and clinical information. Unlike binary classification models that classify only two outcomes, multinomial logistic regression can effectively handle multiple disease classes, making it suitable for real-world healthcare applications.

The proposed system aims to support doctors by providing quick preliminary disease predictions, reducing diagnosis time, and improving decision-making. It also serves as an educational demonstration of applying machine learning techniques to healthcare problems. Although the system does not replace medical professionals, it acts as a decision-support tool that enhances healthcare efficiency and contributes to better patient outcomes.

3. Software Requirements:

a) Development Environment

- Programming Language: Python 3.10 or above
- IDE: Google Colab

b) Machine Learning Tools

- Algorithm: Multinomial Logistic Regression
- Library: Scikit-learn

c) Data Processing

- Libraries: Pandas, NumPy

d) Data Visualization

- Libraries: Matplotlib, Seaborn

e) Dataset

- Source: Kaggle Healthcare Dataset

f) Platform Requirements

- Operating System: Windows 10/11, Linux, or macOS
- Web Browser: Google Chrome or Microsoft Edge

g) Version Control

- Tool: Git and GitHub



Signature of the Guide



Course Instructor